Ethical Case: “A Wave of Trouble”

The Meltdown of Three Nuclear Reactors at the Fukushima Daiichi Power Plant

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Background:

Japan is in the Pacific Ocean to the east of China and is comprised of more than 6,000 islands. On March 11, 2011, a magnitude 9.0 earthquake struck near the coast of the Japanese mainland and caused several aftershocks and large tsunami waves. The 15 feet tall tsunami waves crashed into mainland Japan and caused chaos at every turn. Within the path of tsunami was the Fukushima Daiichi nuclear power plant. The power plant consists of six nuclear reactor units. Units 1, 2, and 3 were operating at full power when it was hit by the tsunami. The nuclear core in unit 4 was unloaded in preparation for refueling. Units 5 and 6 were in a cold shut down condition [4]. Due to a critical design failure, and as a result of a tsunami, the Fukushima Daiichi nuclear power plant experienced nuclear meltdown; a nuclear disaster with its effects still felt today and likely long into the future.

The Problem:

When the earthquake occurred, the power plant's seismic sensors were tripped, causing a sudden insertion of reactor control rods in what is referred to as a "SCRAM", which completely halts the fission reactions taking place in the reactor core. This safety feature worked exactly as designed and placed the reactors in the safest possible condition following a seismic event. However, even after a nuclear reactor is shut down, it continues to produce thermal energy referred to as "decay heat" which results from the continued radiation decay of the existing fission products until they reach a stable isotope. Decay heat ranges from 7% of average reactor power (immediately after shutdown) to about 1.5% [1].

The March 11th earthquake was one of the largest seismic events in the 20th century, and although it did not significantly damage the power plant, it did cause the plant's external power supply to be lost. In the event of the plant's loss of external power, backup diesel generators are designed to come online automatically. The Fukushima Daiichi nuclear power plant had 13 backup diesel generators brought online when it lost its external power [1].

Other safeguards existing at the Fukushima Daiichi nuclear power plant included a 10-meter seawall to protect against tsunamis and the plant itself was built 10 meters above level to protected against flooding. When the plant was designed in 1960, these safeguards were deemed satisfactory. However, during the end of the 21st century, scientific studies began to suggest that the region was much more susceptible to a future tsunami of much greater magnitude and destructiveness. This meant that the safety features deemed satisfactory in 1960 were no longer adequate. Although the scientific community upgraded its tsunamic risk for the area, the plant failed upgrade its tsunami protections [1].

On Marth 11th, when the 15-meter tsunami crashed into Fukushima Daiichi, the lack of its tsunami protections became evident. The tsunami barreled over the plant's seawall and flooded the plant's facilities. Although the plant was built 10 meter above sea level, the plants critical seawater pumps were built only 4 meters above sea level in the basement. The seawater pumps were quickly inundated with water and rendered inoperable. The Fukushima power plant relied on the seawater pumps to help facilitate the heat transfer through the main condenser unit and cooling circuits. Additionally, the electrical switchgear, batteries, instrumentation, control, and lighting were flooded and went offline. Moreover, the backup generators shut down as a result of the flooding [2]. Without electricity and the means to facilitate heat transfer from the reactors, the ability to remove decay heat from the nuclear reactors failed.

Although the fission reaction in the reactors was completely inhibited, thermal power, and thus heat, was still being produced. Heat production requires some form of heat transfer in order to maintain a desired temperature. Without heat transfer from the reactors, the nuclear reactor core temperatures increased to above 2300 °C [1]. The reactor cooling water vaporized, pressures increased in the reactor vessel, and the core material (uranium in a zircoloy cladding) melted into what is referred to as corium. The corium then melted its way downward, with the potential to melt its way through the bottom of the core containment and release an enormous amount of radioactive material to the surrounding environment and water.

When the powerplant lost all means of electricity, it experienced a station blackout and the area around the plant had to be evacuated. Because of the catastrophic damage to the surrounding area from the tsunami, there was no way to restore the plants power supply from the grid. At least 100,000 people had to leave their homes, many of whom can never return. As a direct result of the nuclear disaster at least 1,000 people died. Additionally, without being able to supply power to the area and with people being displaced from the area around the plant, additional deaths piled up, with estimate being around 75,000 [1].

The efforts to control the reactor pressure lead to the venting of steam and hydrogen within the reactor building and resulted in multiple hydrogen explosions. Additionally, in an attempt to control the reactor temperature, seawater was continuously injected to the units without means to contain the radioactive water/vapor. Thus, radioactive material was released into the atmosphere and the ground. Additionally, by introducing seawater into a system only designed to handle distilled water, the is a significant risk of severe corrosion and leaks in the reactor system in the future. Estimates of the contamination level surrounding the plant currently are between 550 and 940 PBq [1].

The Path Forward:

While it is evident that many of the safety features in the reactor design worked as intended, this was insufficient to prevent this disaster. The root cause of this accident is that the only protection in the event of a tsunami was the existing seawall. This is what is often referred to as a single point of failure (SPOF). In a process as potentially unsafe as controlled fission, there should never be a single point of failure when it comes to reactor safety. This critical flaw runs directly counter to the AICHE ethical rule "Hold paramount the safety, health and welfare of the public and protect the environment in performance of their professional duties [3]". In addition to allowing a SPOF to exists in the first place, evidence had previously come to light that the plant should expect tsunamis of much larger size than the original seawall was intended for. In addition to that, there were no mitigative safeguards in place if the seawall was breached or failed altogether. These failures not only violate the AICHE ethical standards, but the standards of the nuclear industry as well.

Moving forward, when deciding to build or continue to operate existing plants in locations like Fukushima Daiichi (susceptible to a natural disaster like a tsunami), it is paramount that these extreme events be thoroughly considered in plant design. At a minimum, it must be shown that no SPOF exist in the protection against major events such as tsunamis. In addition, these types of disasters are not limited to tsunamis, but other destructive events of nature that vary by geographical region. This means that engineers designing these plants cannot just take an existing plant design and assume it will work in another location. In depth analysis of the risks of the location must take place and be incorporated into the reactor protection systems and should be reevaluated on a recurring basis.

The Fukushima nuclear accident points to the need for safer reactor design. The occurrence of the Fukushima nuclear accident in 2011 is undoubtedly a wake-up call about the development of nuclear power throughout world. The various governmental and regulatory bodies that exist globally can not be allowed to fall into complacency and must continually push the companies that operate these plants to regularly reassess the risk posed to the plants. This is especially true now as we enter a period where extreme events are likely to escalate due to global warming and sea level rise.

Conclusion:

The critical element of this accident opens it up for ethical debate is that the company operating the plant as well as the regulatory body for Japan knew that the plant was susceptible to a much larger tsunami than originally thought and did nothing. We can argue about a lack of mitigative safeguards and a SPOF, but at the end of the day people knew that the existing seawall was insufficient and did nothing. This complacency is what runs counter to the AICHE code of ethics, a standard which is accepted worldwide. It’s not unexpected for a company to delay upgrades to save money, but when the regulating body that governs them becomes complacent as well it’s a recipe for disaster. These regulatory bodies exist in order to protect the public against events like 3-mile Island or Chernobyl which have already happened. When we become complacent, we open ourselves up to the exact risks posed by these earlier nuclear accidents. This type of problem is not whether or not we can continue to build nuclear plants, this problem is man-mad. We have tools to protect, properly utilizing them is the dilemma.

References:

1. Fukushima Daiichi Accident. (n.d.). Retrieved November 20, 2019, from https://www.world-nuclear.org/information-library/safety-and-security/safety-of-plants/fukushima-accident.aspx.
2. Acton, J. M., & Hibbs, M. (n.d.). Why Fukushima Was Preventable. Retrieved November 20, 2019, from https://carnegieendowment.org/2012/03/06/why-fukushima-was-preventable-pub-47361.
3. Code of Ethics. (2019, January 14). Retrieved November 20, 2019, from https://www.aiche.org/about/code-ethics.
4. Institut de Radioprotection et de Sûreté Nucléaire – IRSN, https://www.youtube.com/watch?v=YBNFvZ6Vr2U&t=137s